Project Summary:

Sentiment Analysis on Product Reviews Using Multinomial Naive Bayes

Objective:

The goal of this project is to develop a machine learning model that classifies product reviews into three sentiment categories: Positive, Neutral, and Negative. The dataset contains text reviews labeled with their corresponding sentiment, and the aim is to predict the sentiment of new reviews.

Steps and Methodology:

1. **Data Loading and Exploration**:
   - The dataset is loaded from a CSV file containing product reviews and their sentiment labels (`Positive`, `Neutral`, `Negative`).
   - A new column, `new_col`, is created by mapping sentiment labels to numerical values (Positive = 2, Neutral = 1, Negative = 0) to facilitate model training.
2. **Data Preprocessing**:
   - Reviews are split into training and testing sets using `train_test_split` with a 70-30 split.
   - The `CountVectorizer` is used to convert the text data into a matrix of token counts, which is a common preprocessing step for text classification tasks.
3. **Model Training**:
   - A Multinomial Naive Bayes model is trained on the tokenized training data. This algorithm is well-suited for text classification due to its assumption of multinomial distribution for feature vectors.
4. **Model Testing**:
   - The model is tested on the tokenized test data. The model's performance is evaluated using several metrics:
     - **Accuracy**: Measures the proportion of correct predictions over total predictions.
     - **Precision**: Evaluates the accuracy of positive predictions.
     - **Recall**: Measures the ability of the model to find all the relevant cases.
     - **F1 Score**: A harmonic mean of precision and recall, providing a balance between the two metrics.
5. **Prediction Examples**:
   - The model is used to predict sentiments of new sample texts, such as:
     - Positive example: "The product is so good. I will order it next time."
     - Negative example: "The product is worse. It’s small, dirty, and packed poorly."
     - Neutral example: "The product is average."
6. **Results**:
   - The model achieves high performance with accuracy, precision, recall, and F1 score all around 99%, indicating strong predictive capabilities on the test data.

Applications:

This project demonstrates how sentiment analysis can be used in real-world applications, such as:

- Analyzing customer feedback to gauge satisfaction levels.
- Automating the categorization of product reviews to improve customer service.
- Identifying trends in customer sentiment over time.

Tools and Libraries:

- **Pandas**: For data manipulation and analysis.
- **NumPy**: For numerical computations.
- **scikit-learn**: For machine learning model building and evaluation.

The project showcases the effective use of Multinomial Naive Bayes for text classification tasks, particularly sentiment analysis. The high accuracy and strong performance metrics validate the model's ability to classify product reviews accurately, making it a valuable tool for businesses to understand customer sentiments.